F3XP M.A.R.S.

A Level Physics Practice Paper 3

PHYSICS **9702/43**

Paper 4 A Level Structured Questions **August/September 2026**

MARK SCHEME

Maximum Mark: 100

Practice paper

This mark scheme goes with the practice question paper of the same number. It shows the answers and the marking points that the paper was written to test, set out in the layout of the 2025 Cambridge International mark schemes.

Alternative answers that are correct physics should be credited even where they are not listed.

Mark categories

B marks Independent marks that do not depend on other marks. The point must be seen in the answer.

M marks Method marks on which later **A** marks depend. If the **M** mark is not awarded, the **A** marks that follow cannot be awarded either.

C marks Compensatory marks. They can be awarded even if the point is not written down, when later working shows that it must have been known. If a numerical answer is correct, all the **C** marks before it are awarded automatically.

A marks Answer marks. They may depend on an **M** mark or allow a **C** mark by implication.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
( )	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the **context** for an answer.
___	Underlined content must be present in the answer to award the mark.
ecf	Error carried forward: an incorrect earlier answer used correctly later gains the later marks.
allow	Lists other answers that gain the mark.
ignore	Lists content that neither gains nor loses credit.

Question	Answer	Marks
1(a)	gravitational force provides the centripetal force: $GMm / r^2 = mr\omega^2$ **or** $GMm / r^2 = mv^2 / r$	**M1**
	$\omega = 2\pi / T$ **or** $v = 2\pi r / T$, and algebra leading to $T^2 = (4\pi^2 / GM)\, r^3$	**A1**
1(b)(i)	$T = 24.0 \times 3600 = 8.64 \times 10^4$ s and $r^3 = GMT^2 / 4\pi^2$	**C1**
	$r^3 = 6.67 \times 10^{-11} \times 5.98 \times 10^{24} \times (8.64 \times 10^4)^2 / 4\pi^2$ $r = 4.23 \times 10^7$ m	**A1**
1(b)(ii)	orbit of X is not in the plane of the equator / X is not always above the equator	**B1**
	X moves north and south (of the equator) relative to the Earth's surface, so it does not stay above the same point on the Earth's surface a period of 24 hours is necessary but not sufficient: a geostationary orbit must also be equatorial and from west to east	**B1**
1(c)(i)	$mv^2 / r = GMm / r^2$ so $E_K = GMm / 2r$ and total energy = $E_K + E_P = -GMm / 2r$	**C1**
	energy supplied = increase in total energy = $\frac{1}{2}GMm\,(1 / r_1 - 1 / r_2)$ $= \frac{1}{2} \times 6.67 \times 10^{-11} \times 5.98 \times 10^{24} \times 1200 \times [1 / (6.80 \times 10^6) - 1 / (4.23 \times 10^7)]$	**C1**
	$= 2.95 \times 10^{10}$ J the increase in gravitational potential energy alone (5.91×10^{10} J) ignores the decrease in kinetic energy: max. 1 mark	**A1**
1(c)(ii)	$v^2 = GM / r$, so speed and kinetic energy decrease as r increases	**B1**
	gravitational potential energy increases by twice the decrease in kinetic energy, so the total energy increases **or** energy supplied = gain in gravitational potential energy – loss in kinetic energy	**B1**

Question	Answer	Marks
2(a)	any two from: • large number of molecules in random motion • no intermolecular forces (except during collisions) • volume of molecules negligible compared with volume of container / gas • collisions (of molecules with each other and with walls) are elastic • time of collision negligible compared with time between collisions	**B2**
2(b)(i)	straight line through the points, extended to $p = 0$	**B1**
	temperature = –273 °C (allow –260 °C to –290 °C)	**B1**
2(b)(ii)	straight line with twice the pressure at each temperature: p = 110 kPa at 0 °C (and 150 kPa at 100 °C)	**B1**
	line meets $p = 0$ at the same temperature as the original line (–273 °C) p is proportional to NT at constant V, so $p = 0$ at $T = 0$ K whatever the amount of gas	**B1**
2(c)(i)	temperature = $(R_0 – R) / (R_0 – R_{100}) \times 100$ $= (32.0 – 3.73) / (32.0 – 0.770) \times 100$	**C1**
	= 91 °C	**A1**
2(c)(ii)	resistance of the thermistor does not vary linearly with (thermodynamic) temperature, so the two scales agree only at the fixed points (where they were calibrated)	**B1**
	the gas thermometer reading (50 °C / 323 K) is the thermodynamic temperature, since p is proportional to T for an ideal gas at constant volume	**B1**
3(a)	sum of (random) kinetic and potential energies	**B1**
	of the molecules	**B1**

Question	Answer	Marks
3(b)	A → B: ΔU = –300 J B → C: W = 0 and ΔU = +300 J	**B1**
	C → A: W = –400 J (area under line, negative because the gas expands)	**B1**
	C → A: ΔU = 0 (ΔU for the whole cycle is zero) and q = +400 J	**B1**
3(c)	temperature is not constant	**B1**
	pV (proportional to T) is 300 J at C and at A but is greater between them, e.g. 400 J at $V = 2.0 \times 10^{-3}$ m^3, so temperature rises and then falls ΔU = 0 means only that the temperatures at C and A are equal	**B1**
3(d)	no work is done (W = 0) because the gas expands into a vacuum / does not push against anything	**B1**
	no thermal energy is transferred (q = 0) because the container is insulated	**B1**
	so ΔU = 0 and, since the internal energy of an ideal gas is kinetic energy of the molecules only, the temperature does not change	**B1**
4(a)	acceleration (directly) proportional to displacement (from a fixed point)	**B1**
	acceleration in the opposite direction to the displacement / directed towards the fixed point	**B1**
4(b)(i)	$E_P = \frac{1}{2}kx^2$ with E_P = 40 mJ at x = 4.0 cm: $k = 2 \times 0.040 / 0.040^2 = 50$ N m^{-1}	**B1**
4(b)(ii)	T: horizontal line at 40 mJ from x = –4.0 cm to x = +4.0 cm	**B1**
	K: inverted parabola, 40 mJ at x = 0 and zero at x = ±4.0 cm, crossing the E_P line at 20 mJ (x = ±2.8 cm)	**B1**
4(c)	same tension in each spring so each extends by half the displacement: effective spring constant = 25 N m^{-1}	**C1**

Question	Answer	Marks
4(c)	$a = -(k / m)x$ so $\omega = \sqrt{(25 / 0.50)} = 7.1\ \text{rad s}^{-1}$	**C1**
	$T = 2\pi / \omega = 0.89$ s the period does not depend on the amplitude, so the 2.0 cm displacement is not used springs treated as in parallel ($k = 100\ \text{N m}^{-1}$, $T = 0.44$ s): max. 1 mark	**A1**
4(d)	energy is proportional to (amplitude)^2 / $E = \frac{1}{2}kx_0^2$: energy remaining = 40 / 4 = 10 mJ	**C1**
	energy dissipated = 40 – 10 = 30 mJ 20 mJ (energy taken as proportional to amplitude) scores 0	**A1**
5(a)(i)	arrow at P perpendicular to the equipotential line, pointing away from the sphere (towards lower potential)	**B1**
5(a)(ii)	zero	**B1**
	P and Q are at the same potential, so work done = $q\Delta V = 0$ (whatever the path)	**B1**
5(b)(i)	field strength = (–) potential gradient / $E = -\text{d}V / \text{d}x$	**B1**
	gradient of the graph at $x = 0$ is not zero (about $90\ \text{V cm}^{-1} = 9 \times 10^3\ \text{V m}^{-1}$), so the field strength at X is not zero	**B1**
5(b)(ii)	tangent at $x = 2.0$ cm: field strength = gradient = $1.5 \times 10^4\ \text{V m}^{-1}$ (allow 1.3×10^4 to $1.7 \times 10^4\ \text{V m}^{-1}$)	**C1**
	$qE = mg$ so $q = 7.2 \times 10^{-16} \times 9.81 / 1.5 \times 10^4 = 4.7 \times 10^{-19}$ C	**C1**
	number = $q / e = 3$ the final answer must be an integer: a non-integer answer (e.g. 2.9) scores max. 2	**A1**
5(b)(iii)	potential and field strength at every point double, so the electric force on the droplet becomes $2mg$ (upwards)	**B1**

Question	Answer	Marks
5(b)(iii)	resultant force = mg upwards so acceleration = 9.81 m s^{-2} upwards	**B1**
6(a)	(magnitude of) charge on one plate per unit potential difference between the plates	**B1**
6(b)(i)	experiment 1: charge 0.5 and p.d. 1 (unchanged)	**B1**
	experiment 2: charge 1 (unchanged) and p.d. 2	**B1**
	energy stored: 0.5 in experiment 1 and 2 in experiment 2	**B1**
6(b)(ii)	charge on the plates decreases, so charge flows back through the battery (against its e.m.f.) / the battery is charged	**B1**
	energy is transferred to the battery: both the work done in pulling the plates apart and the decrease in stored energy are transferred to the battery	**B1**
6(c)(i)	gradient = –1 / RC gradient = –(2.30 – 0.18) / 10 = –0.21 s^{-1}	**C1**
	R = 1 / (0.213 × 470 × 10^{-6}) = 1.0 × 10^4 Ω	**A1**
6(c)(ii)	energy is proportional to V^2, so energy halves when $V = V_0 / \sqrt{2}$ / when ln (V / V) has decreased by ½ ln 2 = 0.35	**C1**
	t = ½RC ln 2 = 0.5 × 4.7 × 0.693 = 1.6 s 3.3 s (the time for the p.d. to halve) scores 0	**A1**
7(a)	e.m.f. induced is proportional to	**M1**
	rate of change of magnetic flux linkage / rate of cutting of magnetic flux	**A1**
7(b)(i)	E = 0 at $t \approx$ 100 ms (when the flux linkage is maximum), with pulses of opposite sign before and after this time	**B1**

Question	Answer	Marks
7(b)(i)	magnitude of E greatest where the gradient of Fig. 7.2 is greatest (at $t \approx 90$ ms and $t \approx 115$ ms) and $E \approx 0$ at $t = 0$ and at $t = 150$ ms	**B1**
	second pulse has a larger peak and a shorter duration than the first $E = -\mathrm{d}(N\Phi) / \mathrm{d}t$, so the first pulse is negative and the second positive; allow the inverted sketch	**B1**
7(b)(ii)	induced current opposes the change that produces it / opposes the motion of the magnet / opposes the decrease in flux linkage	**B1**
	force on the magnet is upwards (the coil attracts the magnet as it leaves) a downward force (magnet "pushed out" of the coil) scores 0	**B1**
7(c)	Hall voltage is reversed in polarity / the other face (Q instead of P, or P instead of Q) is now at the higher potential	**B1**
	positive carriers move in the opposite direction to electrons, so the magnetic force (BQv) on them is in the same direction: they collect on the same face, but the charge there is now positive	**B1**
	Hall voltage is four times larger because $V_H = BI / ntq$, so V_H is proportional to $1 / n$ (the drift speed is four times larger)	**B1**
8(a)	the value of the direct current	**B1**
	that dissipates (thermal) energy at the same (average) rate in a resistor	**B1**
8(b)(i)	D2 and D3	**B1**
8(b)(ii)	full-wave rectified p.d. has the same r.m.s. value as the supply: $V_{r.m.s.} = 17.0 / \sqrt{2} = 12.0$ V	**C1**
	$P = V_{r.m.s.}^2 / R = 12.0^2 / 150 = 0.96$ W allow $P = V_0^2 / 2R$	**A1**

Question	Answer	Marks
8(c)(i)	half-wave output: the humps between 0 and 10 ms and between 20 ms and 30 ms only (peak 17 V), with V = 0 from 10 ms to 20 ms	**B1**
8(c)(ii)	mean power / mean value of V^2 is halved	**C1**
	$V_{r.m.s.}$ = 12.0 / √2 (= 17.0 / 2) = 8.5 V 6.0 V (r.m.s. value halved) scores 0	**A1**
8(d)	ripple is greater / minimum p.d. is lower	**B1**
	peaks occur every 20 ms instead of every 10 ms, so the capacitor discharges (through the resistor) for about twice as long between peaks	**B1**
9(a)(i)	$p = \sqrt{(2m_e eV)} = \sqrt{(2 \times 9.11 \times 10^{-31} \times 1.60 \times 10^{-19} \times 4000)} = 3.4 \times 10^{-23}$ N s	**C1**
	$\lambda = h / p = 6.63 \times 10^{-34} / 3.4 \times 10^{-23} = 1.9 \times 10^{-11}$ m	**A1**
9(a)(ii)	p is proportional to $\sqrt{V}$ so λ is proportional to $1 / \sqrt{V}$: the wavelength halves (it does not decrease to one quarter)	**B1**
	angle of diffraction (approximately) proportional to wavelength, so the radius of each ring (approximately) halves	**B1**
9(b)(i)	6	**B1**
9(b)(ii)	longest wavelength from the smallest energy difference: –3.7 – (–5.5) = 1.8 eV	**C1**
	$\lambda = hc / \Delta E = (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / (1.8 \times 1.60 \times 10^{-19}) = 6.9 \times 10^{-7}$ m the 2.1 eV transition between the two highest levels (5.9×10^{-7} m) scores 0	**A1**
9(c)(i)	binding energy of products = (141 × 8.33) + (92 × 8.51) = 1957 MeV binding energy of uranium-235 = 235 × 7.59 = 1784 MeV	**C1**

Question	Answer	Marks
9(c)(i)	energy released = 1957 – 1784 = 174 MeV free neutrons have no binding energy	**A1**
9(c)(ii)	binding energy per nucleon increases with nucleon number for light nuclei, so the product nucleus has a greater (total) binding energy (and less mass) than the two light nuclei	**B1**
10(a)(i)	(total) momentum of the positron and electron before annihilation is (approximately) zero	**B1**
	momentum is conserved, so the photons have equal and opposite momenta / a single photon would have non-zero momentum	**B1**
10(a)(ii)	difference in distances travelled = $c\Delta t = 3.00 \times 10^{8} \times 0.40 \times 10^{-9} = 0.12$ m	**C1**
	distance from M = $c\Delta t / 2 = 0.060$ m (on the D_1 side of M) 0.12 m scores 1 mark	**A1**
10(b)(i)	$\Delta\lambda / \lambda = v / c$, where λ is the emitted (laboratory) wavelength	**C1**
	$v = 3.00 \times 10^{8} \times (689.1 - 656.3) / 656.3 = 1.50 \times 10^{7}$ m s^{-1} 689.1 nm in the denominator (1.43×10^{7} m s^{-1}) scores 1 mark	**A1**
10(b)(ii)	$d = v / H_0 = 1.50 \times 10^{7} / 2.3 \times 10^{-18} = 6.5 \times 10^{24}$ m	**B1**
10(b)(iii)	time = $d / v = 1 / H_0 = 4.3 \times 10^{17}$ s	**C1**
	age = $4.3 \times 10^{17} / (3.16 \times 10^{7}) = 1.4 \times 10^{10}$ years the speed and distance of the galaxy cancel: any galaxy that obeys Hubble's law gives the same estimate	**A1**